DOCTORATE IN COMPUTER SCIENCE AND ENGINEERING

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Use of Differential Privacy for Unobservable Privacy-Preserved Communication
Some thoughts on the Life, the Universe, and Everything Else

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Dedicatory lorem ipsum.

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Acknowledgments are personal text and should be a free expression of the author.

However, without any intention of conditioning the form or content of this text, I would like to add that it usually starts with academic thanks (instructors, etc.); then institutional thanks (Research Center, Department, Faculty, University, FCT / MEC scholarships, etc.) and, finally, the personal ones (friends, family, etc.).

But I insist that there are no fixed rules for this text, and it must, above all, express what the author feels.

"You cannot teach a man anything; you can only help him discover it in himself."

— **Galileo**, Somewhere in a book or speach
(Astronomer, physicist and engineer)

Abstract

We connect to the internet daily, gaining access to an overwhelming flow of information about the world. However, some of the most dramatic and impactful news—especially rare reports from regions often absent from our usual feeds—rely on anonymous sources. These reports frequently emerge from places known for limited freedom of speech and oppressive conditions, making their very existence both precarious and vital. This information come from individuals who risk their lives to share their stories, often relying on anonymity networks to protect their identities and ensure their safety. With the vast resources and power of the prosecutors, often totalitarian regimes, need for robust, secure and privacy-preserving is more critical than ever.

To tackle this challenge, we aim to develop a communication network, on top of Tor network, capable of countering state-of-the-art correlation attacks to ensure user anonymity and privacy, all while maintaining practical levels of throughput and latency. By integrating differential privacy principles, this work aims to inject calibrated noise into traffic flows and statistical communication patterns, maintaining high privacy guarantees, even under advanced traffic analysis and censorship attacks.

Keywords: Anonymity Networks, Differential Privacy, Privacy-Preserving Communication, Covert Communication, WebRTC, Privacy Leakage, Traffic Analysis Attacks

Resumo

Ligamo-nos diariamente à Internet e temos acesso a uma avassaladora quantidade de informações sobre o mundo. No entanto, algumas das notícias mais dramáticas e impactantes, especialmente os relatos raros de regiões frequentemente ausentes dos nossos 'feeds' habituais, dependem de fontes anónimas. Estes relatos surgem frequentemente em locais conhecidos pela liberdade de expressão limitada e por condições opressivas, o que torna a sua existência precária e vital. Estas informações provêm de indivíduos que arriscam a vida para partilhar as suas histórias, recorrendo frequentemente a redes de anonimato para proteger as suas identidades e garantir a sua segurança. Com os vastos recursos e o poder dos procuradores, muitas vezes regimes totalitários, a necessidade de um sistema robusto, seguro e que preserve a privacidade é mais crítica do que nunca.

Para fazer face a este desafio, pretendemos desenvolver uma rede de comunicações, sobre a rede Tor, capaz de contrariar ataques de correlação de última geração para garantir o anonimato e a privacidade dos utilizadores, mantendo simultaneamente níveis práticos de 'throughput' e latência. Ao integrar princípios de 'Differential Privacy', este trabalho visa injetar ruído calibrado nos fluxos de tráfego e nos padrões estatísticos de comunicação, mantendo elevadas garantias de privacidade, mesmo sob ataques avançados de análise de tráfego e de censura.

Palavras-chave: Redes de Anonimato, Privacidade Diferencial, Comunicação Preservadora de Privacidade, Comunicação Encoberta, WebRTC, Fuga de Privacidade, Ataques de Análise de Tráfego

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1 Introduction

The internet has become an integral part of the daily life for most of the world's population. However, in some regions, individuals are denied the ability to freely share their stories or access vital information due to oppressive conditions and restrictions on freedom of speech. This oppression and restrictions are often enforced by totalitarian regimes that have vast resources and power to control the flow of information. [0, 0, 0]

Anonymity networks have become essential tools for journalists and whistleblowers in regions with oppressive regimes, enabling them to communicate privately and securely. In response, these regimes have intensified their efforts to undermine and suppress the use of such networks, employing advanced censorship techniques and surveillance strategies to restrict access and discourage their adoption. [**ADD WHAT MEASURES HAVE BEEN MADE**]

Tor, one of the most popular anonymity networks, is an anonymous communication service based on the Onion Routing Protocol. [0] Tor is designed to protect users' privacy and security by routing their traffic through a network of volunteer-operated servers, encrypting it at each step, and ensuring that no single server knows both the source and destination of the traffic.

1.1 Motivation

Anonymity networks have become increasingly indispensable as they provide critical tools for ensuring privacy and security in digital communications. They are particularly vital for journalists, whistleblowers, and ordinary users operating under oppressive regimes, where freedom of expression is severely restricted, and surveillance is pervasive. These networks empower individuals to share information safely, evade censorship, and protect their identities, thereby upholding the principles of free speech and press independence in challenging environments.

To enforce and improve Tor's privacy and attack resistance, researchers have proposed various solutions[ADD SOME PAPERS ABOUT BRIDGES/TOR RELAY NODES/TOR-ONION-SERVICES/TOR HIDDEN SERVICES]. More recently, other solutions have

emerged, such as Tor-strengthening mechanisms based on K-anonymity input covert channels [Horta 22 + J.Teixeira MA THESIS FCT/UNL], TorKameleon that employes K-anonymization techniques and covert WebRTC-based channels traffic encapsulation [TORKAMELEON REF],

Differential Privacy is a privacy-preserving technique that introduces controlled noise into data and query results, preventing sensitive information from being exposed. This controlled noise can be parameterized and adjusted to provide a trade-off between privacy and utility. In this context, the formal proofs that Differential Privacy provides can be used to ensure privacy rigor and provide higher confidence against attacks.

1.2 Problem

Although anonymity networks are designed to provide robust privacy and security, advancements in traffic analysis and correlation techniques have increasingly exposed vulnerabilities, reducing their reliability and trustworthiness for users—particularly in regions where pervasive surveillance by authoritarian regimes is a critical concern. These regimes actively invest in sophisticated surveillance methods and traffic manipulation strategies to maintain control over digital communication.

Tor, despite being one of the most widely used anonymity networks, is not immune to these challenges. It remains vulnerable to several attack vectors, including traffic correlation, congestion-based attacks, timing-based correlations, and fingerprinting techniques. [0, 0, 0, 0] Such vulnerabilities highlight the ongoing arms race between those striving to safeguard online anonymity and those seeking to undermine it through technical and infrastructural advancements.

The absence of standardized privacy metrics for evaluating privacy guarantees in a formal and rigorous manner raises concerns about the effectiveness of anonymizing networks like Tor. Research has demonstrated vulnerabilities in the Tor network to various attacks, including traffic analysis. For instance, Chakravarty et al. showed that statistical correlation techniques can be used to perform successful traffic analysis attacks on the Tor network. Additionally, Shi and Matsuura demonstrated that user privacy can be compromised through fingerprinting attacks.

1.3 Goals

The goal is to add differential privacy (DP) mechanisms into the Tor open-source project to preserve its existing security guarantees while enhancing its privacy protections. By incorporating DP, the aim is to provide quantifiable and adjustable privacy guarantees to users, addressing vulnerabilities to correlation attacks and improving resistance to traffic observability. Furthermore, the project seeks to analyze the trade-offs between privacy and utility within the Tor context and to evaluate the performance of the proposed solution compared to the original Tor network.

As stated in Section, it is important to ensure that anonymity networks are secure against threats like artificial intelligence and machine learning, and to address the lack of standardized privacy metrics to evaluate the performance of anonymity networks. We aim to create and implement a new algorithm that can be used in anonymity networks that leverages differential privacy to ensure privacy and security with higher performance than existing solutions. We also aim to evaluate the performance of the algorithm and compare it with other anonymity networks to prove that this solution is a viable alternative to usual anonymity networks solutions. To test the algorithm and its performance, we intend to realize correlation attacks on the existing networks and on our prototype and compare the results. To test performance, we will measure the latency and throughput of the algorithm and compare it with other anonymity networks.

1.3.1 Contributions

This thesis tries to address the problem of ensuring privacy and security in anonymity networks while maintaining high performance. In particular, the contributions of this thesis are:

1. Design a new Differential Privacy Based Algorithm for Privacy Preserved Communication in Anonymity Networks.

2. Implement the algorithm on a real-world scenario, develop a prototype and provide a proof of concept;

3. Evaluate the performance of the algorithm and compare it with other anonymity networks; This contribution aims to prove if this solution is a viable alternative to usual anonymity networks solutions and if, as mention before, can allow users to switch from traditional to anonymity networks.

1.4 Report Organization

The remainder of this report is organized as follows: Chapter introduces anonymity networks and differential privacy, as well as some other related topics. The proposal of work is presented on Chapter, which provides some system model and guidelines for the development and discusses the proof method to verify the work. Finally, workplan is explained and addressed in Chapter

2

Related Work

2.1 Tor Network

Tor is a circuit-based low-latency anonymous communication service based on *onion routing* for asynchronous, loosely federated onion routers [Tor: The second-generation onion router]. This protocol is a low-latency anonymity network design that tries to protect users from correlation attacks. The overlay network relies on *Onion Routers* (ORs) that maintain TLS connection to every other onion router.

Onion Proxy are local software, run on the client and handle connections with the users' applications, fetch the current network information and the list of *Onion Routers* (ORs) and build circuits through the network. Each router maintains TLS connections to nodes they have been in contact recently and a pair of keys: long-term identity key and short-term onion key. Long-term keys are used to sign TLS certificates and its router descriptor. Short-term keys are used to decrypt requests, negotiate ephemeral keys and set up circuits.

Traffic travels the network through 512 bytes fixed-size cells with a header and a payload. The header contains the circuit identifier, the command to describe what to do with the payload. Based on the header's command, cells are either control cells or relay cells. Control cells are used to manage circuits and connections, while relay cells are used to carry data.

The sequences of ORs which traffic is routed through are called *circuits*. The user's Onion Proxy construct circuits, negotiating a symmetric key with each OR on the circuit, one hop at a time. Circuits incrementally extends, as well as torn down. Relay cells can only be sent after a circuit is established. Once established, relay cells are encapsulated in several layers of encryption, correspondent to each node in the circuit, with the pre-negotiated keys with each router. As it passes through each node, a layer of encryption is peeled off and forward to the next node. This ensures that only the last node knows the final destination of the data, and no single node knows both the source and destination of the data.

2.2 Anonymity Networks & Mixnets

2.3 K-anonymity

2.4 TorKameleon & Stream Covered Channels

2.5 Differential Privacy

2.6 Summary

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Elaboration

3.1 System Model

3.2 Prototype Guidelines

3.3 Validation

3.4 Summary

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Workplan

Bibliography

[0] M. Alimardani and S. Milan. "The Internet as a Global/Local Site of Contestation: The Case of Iran". In: *Global Cultures of Contestion*. Ed. by D. T. Della and G. K. Thomas. Springer, 2018, pp. 171–192. URL: https://link.springer.com/chapter/10.1007/978-3-319-78889-7_9.

[0] S. Aryan, H. Aryan, and J. A. Halderman. "Internet Censorship in Iran: A First Look". In: *Proceedings of the 3rd USENIX Workshop on Free and Open Communications on the Internet (FOCI 13)*. USENIX Association, 2013. URL: https://www.usenix.org/conference/foci13/workshop-program/presentation/aryan.

[0] S. Chakravarty et al. "On the Effectiveness of Traffic Analysis against Anonymity Networks Using Flow Records". In: *Passive and Active Measurement*. Ed. by M. Faloutsos and A. Kuzmanovic. Cham: Springer International Publishing, 2014, pp. 247–257.

[0] R. Dingledine, N. Mathewson, and P. Syverson. "Tor: The Second-Generation Onion Router". In: *Proceedings of the 13th USENIX Security Symposium*. USENIX Association, 2004, pp. 303–320. URL: https://www.usenix.org/conference/usenixsecurity04/tor-second-generation-onion-router.

[0] R. Jansen, T. Vaidya, and M. Sherr. "Point Break: A Study of Bandwidth Denial-of-Service Attacks against Tor". In: *28th USENIX Security Symposium (USENIX Security 19)*. Santa Clara, CA: USENIX Association, 2019-08, pp. 1823–1840. ISBN: 978-1-939133-06-9. URL: https://www.usenix.org/conference/usenixsecurity19/presentation/jansen.

[0] A. Johnson et al. "Users Get Routed: Traffic Correlation on Tor by Realistic Adversaries". In: *Proceedings of the ACM SIGSAC Conference on Computer and Communications Security (CCS)*. ACM, 2013. DOI: 10.1145/2508859.2516651. URL: https://doi.org/10.1145/2508859.2516651.

[0] J. M. Lourenço. *The NOVAthesis LaTeX Template User's Manual*. NOVA University Lisbon. 2021. URL: https://github.com/joaomlourenco/novathesis/raw/main/template.pdf.

[0] Y. Shi and K. Matsuura. "Fingerprinting Attack on the Tor Anonymity System". In: *Information and Communications Security*. Ed. by S. Qing, C. J. Mitchell, and G. Wang. Berlin, Heidelberg: Springer Berlin Heidelberg, 2009, pp. 425–438.

[0] P. Winter and S. Lindskog. "How the Great Firewall of China is Blocking Tor". In: *Proceedings of the 2nd USENIX Workshop on Free and Open Communications on the Internet (FOCI)*. USENIX Association, 2012. URL: https://www.usenix.org/conference/foci12/workshop-program/presentation/winter.

[0] J. L. Zittrain et al. "The Shifting Landscape of Global Internet Censorship". In: Accessed: 2024-11-17. Berkman Klein Center for Internet Society, 2017. URL: https://www.ssrn.com/abstract=2993485.